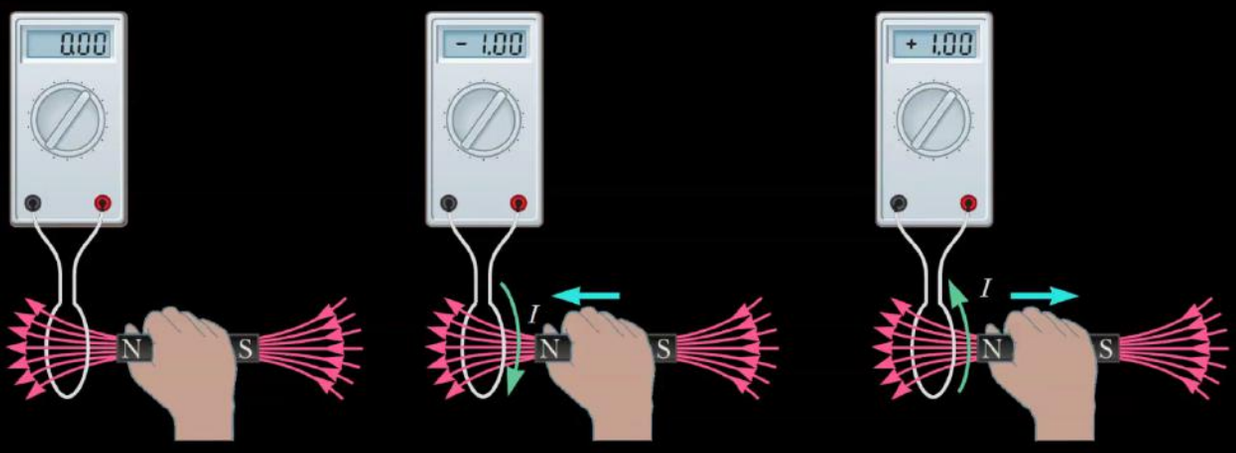
Induced Emf Experiments:



Moving Magnet Relative To Coil Produces Emf



(a) Moving Smaller Current Carrying Coil Produces Emf
(b) Changing Current In The Smaller Coil Produces Induced Emf.

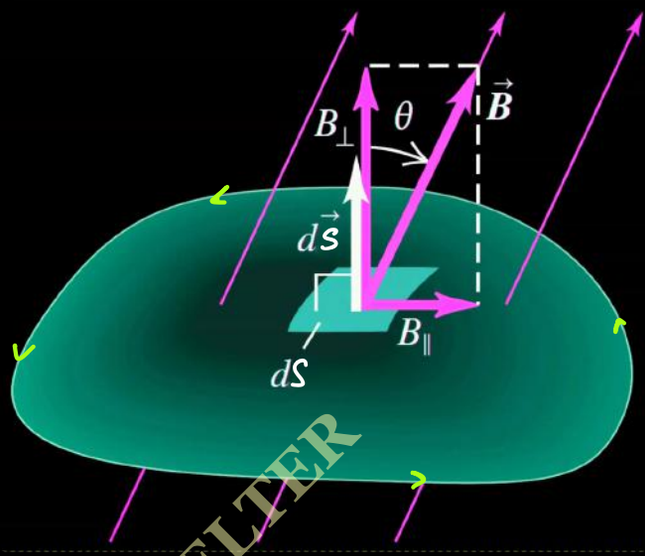
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After oersted's experiment, several experiments were done trying to relate electricity and magnetism, some of which are shown above. It was found that an EMF was induced in the coil if:
(a) Strength of magnetic field increased or decreased
(b) Coil orientation was changed
(c) Area of coil was changed
(d) Number of turns was increased or decreased
(e) Faster the changes greater was the induced EMF
Common phenomenon in all of these experiments was that an EMF was induced whenever magnetic flux through the coil changed, this led to formulation of Faraday's law which we will state in

a following Section.

Magnetic Flux Through A Directed Loop:

A directed loop is one to whose perimeter we have assigned a sense of traversal (clockwise or anticlockwise for planar loops). Once a sense of traversal is specified, we can specify direction of area normal for a surface mounted on the perimeter. For finding area normal, we curl the fingers along the loop, now thumb gives the side on which normal is to be chosen. Now let $d\vec{S}$ be the area vector associated with a patch element mounted on periphery of the loop then we define magnetic flux of the patch as:



$$d\phi = \vec{B}\cdot d\vec{S}$$ thus net flux is $$\phi = \int \vec{B}\cdot d\vec{S}$$

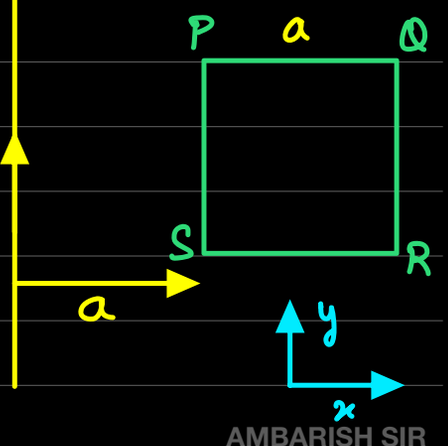
→ For convenience try to (chose the traversal direction of the loop so that area normal makes an acute angle with magnetic field (thus making magnetic flux positive). We will be needing to find the changes in magnetic flux and it is generally easier on the mind dealing with positive quantities.

→ Just like electric flux, magnetic flux also depends on number of magnetic field lines crossing a loop. If the number of field lines crossing is same for two loops then magnetic flux is same.

Example: A square loop is placed near a current carrying infinite wire as shown.

(a) Find the magnetic flux through the loop (choose normal suitably to keep the flux positive)

(b) Find the magnetic flux if the the loop were rotated about line PS by a 90° angle with $\vec{\omega}$ along $\hat{j}$.

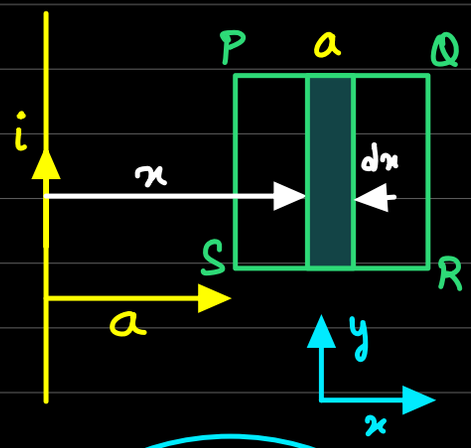


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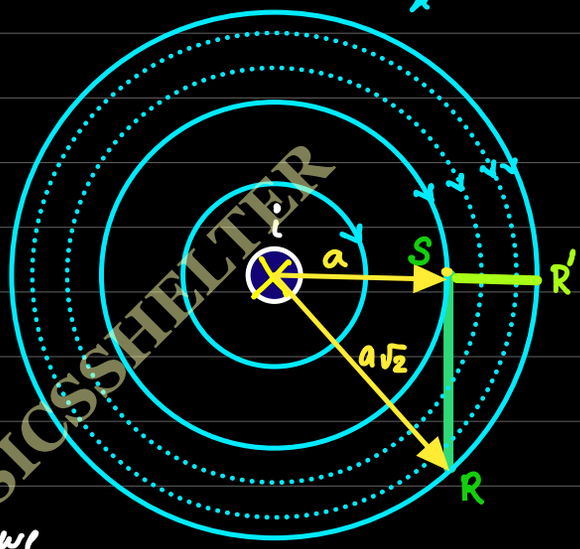
Solution: Consider a small element of the loop as shown, we consider inward direction of normal as positive now:

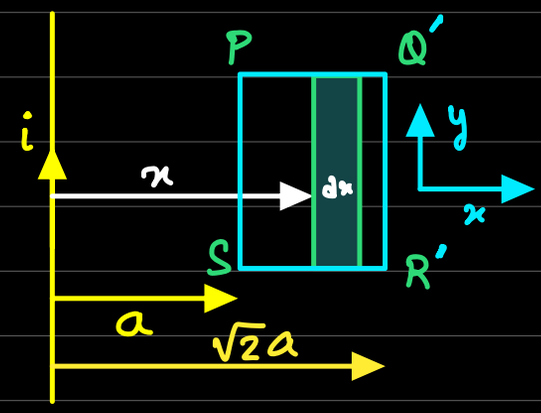
$$d\phi = \frac{\mu_0 i}{2\pi x} \times a\,dx \Rightarrow \phi = \frac{\mu_0 i a}{2\pi}\int_a^{2a} \frac{1}{x}\,dx$$

$$\Rightarrow \phi = \frac{\mu_0 i a}{2\pi}\ln(2) \text{ wbm.}$$



(b) Now when loop is rotated through 90°, we draw a view where the current appears to go in to the page, in this projection looks like a line whose nearest edge is at a distance of a and farthest edge at a distance of $a\sqrt{2}$. If we imagine a a coplanar loop with the wire (SR' in projection), with inward dimensions same as the SR loop, we see that such a loop has same number of field lines as our loop. Thus we can find through the loop SR' to evaluate our required flux. Now converting this problem in our original projection we get the adjacent figure. Thus, $d\phi = \frac{\mu_0 i}{2\pi x}\,a\,dx$





Thus $\phi = \int_a^{\sqrt{2}a} \frac{\mu_0 i}{2\pi x}\,a\,dx = \frac{\mu_0 i a}{2\pi}\ln(\sqrt{2}) = \frac{\mu_0 i a}{4\pi}\ln(2)$ wbm.

(Flux is half of the previous case).

FARADAY'S LAW: The EMF induced in any loop or coil is given as

$$\boxed{\mathcal{E} = \frac{-d\phi}{dt}}$$ and if the coil has N turns, $$\boxed{\mathcal{E} = -N\frac{d\phi}{dt}}$$

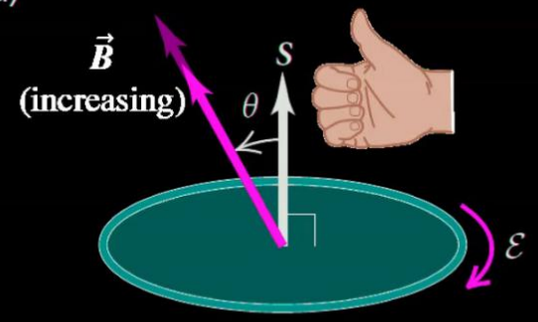
→ Here − sign signifies the sense of the induced EMF. If induced EMF comes out to be +ve then EMF is in the same sense as the sense of the directed loop and if it is −ve, the EMF

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is opposite to the sense of the loop. (Sense of the loop is given by right hand thumb rule, i.e keep thumb along the area normal and curl of fingers of right hand gives the sense of the loop).

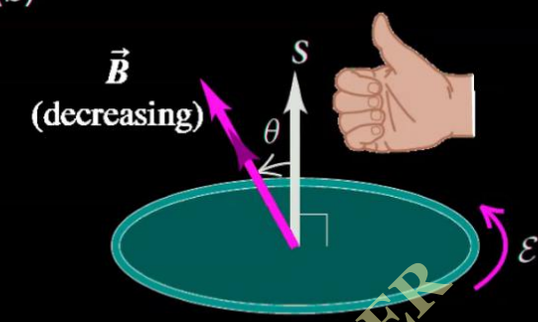
→ To understand this properly understand the following cases:

(a)



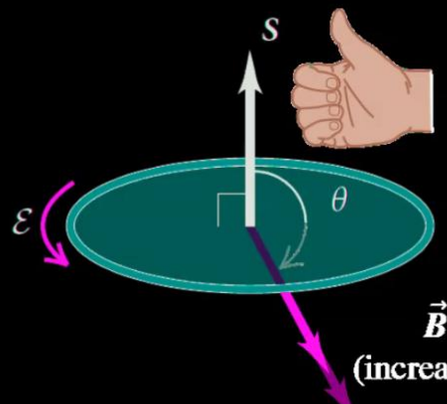
- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming more positive ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

(b)



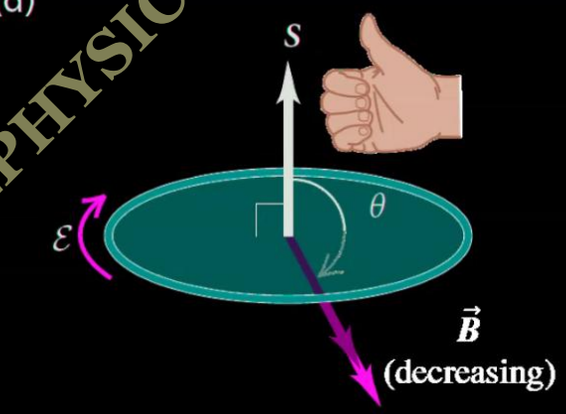
- Flux is positive ($\Phi_B > 0$) ...
- ... and becoming less positive ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).

(c)



- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming more negative ($d\Phi_B/dt < 0$).
- Induced emf is positive ($\mathcal{E} > 0$).

(d)



- Flux is negative ($\Phi_B < 0$) ...
- ... and becoming less negative ($d\Phi_B/dt > 0$).
- Induced emf is negative ($\mathcal{E} < 0$).

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MEANING OF EMF IN A LOOP: Recall that integration of electrostatic force in a closed loop is zero thus in a complete loop work done by electrostatic force must be zero. However, if current is flowing in a loop having some resistance, heat will be getting dissipated and therefore some net work must be done on the charges in the loop at the expense of which this heat is getting evolved. Thus we need some non-conservative force in the loop responsible for pushing the charge and doing net work on charges. The non conservative force per unit charge is denoted by $\vec{E}^*$ since it has same

dimension as that of electric field (like Electric field, this is also a force per unit charge). Now the EMF is defined as work done by this non conservative force per unit charge. Thus.

→ In any portion of the loop (say from A to B) is given as:

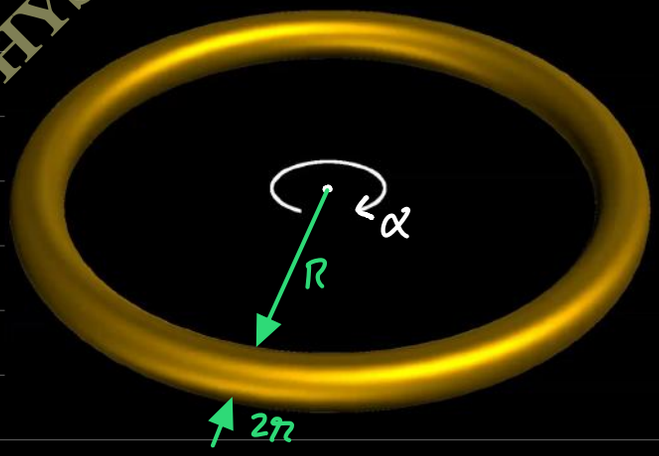
$$\mathcal{E} = \int_A^B \vec{E}^* \cdot d\vec{l}$$

→ In any closed loop, EMF is given as $\oint \vec{E}^* \cdot d\vec{l}$ where circle on the integral sign signifies integration over the entire loop.

→ $\vec{E}^*$ could be of various origins such as magnetic forces, pseudo force, non-electrostatic electric force and chemical force etc.

EXAMPLE: A gold ring of radius R and cross sectional radius r ($r << R$) rotates with an angular acceleration α. Resistivity of the ring is ρ from the ring frame calculate:

(a) E^* (b) EMF in the loop with proper sense (c) Current in the loop (d) Power generated in the loop.



Solution: Here in the ring frame, EMF is generated due to euler force (pseudo) in the tangential direction, thus $E^* = mR\alpha/e$ where m is mass of electron and e is the electron charge, this is in same sense as α since electron has a negative charge, thus EMF is in clockwise sense (even though pseudo force is anticlockwise). Thus $E^* = mR\alpha/e$ Ans.

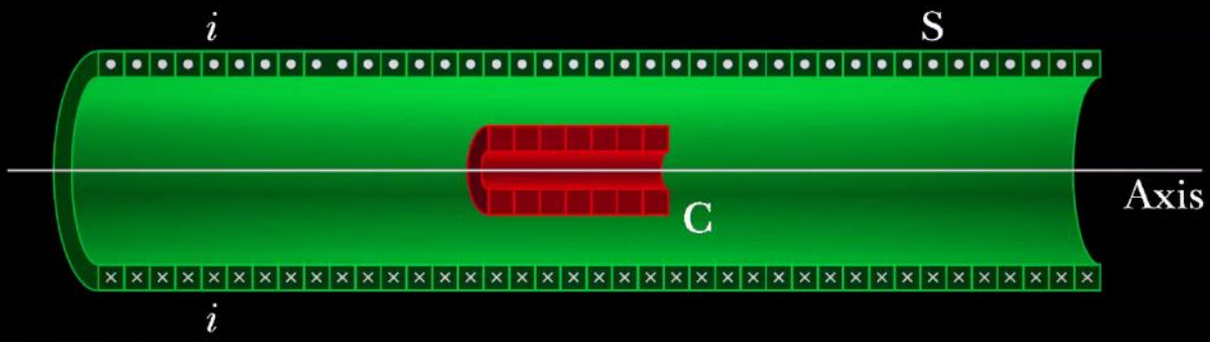
(b) $\mathcal{E}MF = \int \vec{E}^* \cdot d\vec{l} = E^* \times 2\pi R = 2\pi mR^2\alpha/e$ Ans.

(c) $I = \mathcal{E}/\text{Resistance} = \mathcal{E}/[\rho \times 2\pi R/\pi r^2] = \mathcal{E}r^2/2\rho R = \frac{\pi}{\rho e} mRr^2\alpha$ Ans.

(d) Now P can be simply calculated as $I^2 \times$ Resistance or $\mathcal{E}I$

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Example :



The long solenoid S shown (in cross section) in Fig. has 220 turns/cm and carries a current i = 1.5 A; its diameter D is 3.2 cm. At its center we place a 130-turn closely packed coil C of diameter d = 2.1 cm. The current in the solenoid is reduced to zero at a steady rate in 25ms. What is the magnitude of the emf that is induced in coil C while the current in the solenoid is changing?

Solution: B due to outer coil: $B = \mu_0 n_1 i$ (n_1 is no. of turns/length of outer coil). Now for second coil flux linkage: $\phi = N_2 \times B \times \pi d^2/4$

Thus: $\phi = \mu_0 n_1 N_2 i \pi d^2/4 \Rightarrow \mathcal{E} = d\phi/dt = \mu_0 n_1 N_2 \pi \frac{d^2}{4} \cdot \frac{di}{dt}$

Putting the values, $\mathcal{E} = 4\pi \times 10^{-7} \times 220 \times 10^2 \times 130 \times \pi \times \frac{0.021^2}{4} \times \frac{1.5}{25 \times 10^{-3}}$ V

$\Rightarrow \mathcal{E} = 75$ mV Ans.

Relation Between Total Charge Flown And Resistance In A Coil :

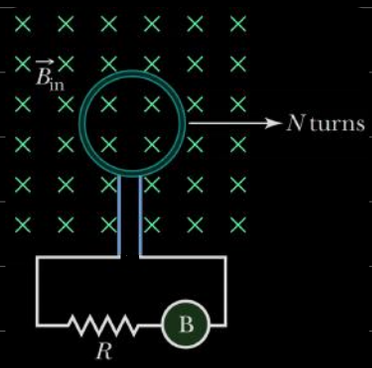
Recall that $\mathcal{E} = -d\phi/dt = R\, dq/dt \Rightarrow dq = -\frac{1}{R} d\phi$

$\Rightarrow$ $\boxed{\Delta Q = \frac{-1}{R} \Delta\phi}$

→ Thus to find total charge flown, we just need to know the total change in flux, it does not matter how that change is brought about.

Example :

A small coil of N turns has its plane perpendicular to a uniform magnetic field as shown in Figure. The coil is connected to a Ballistic galvanometer, a device designed to measure the total charge passing through it. Find the charge passing through the coil if the coil is rotated through 180° about its diameter. Area of each turn is A, Resistance of coil is R.



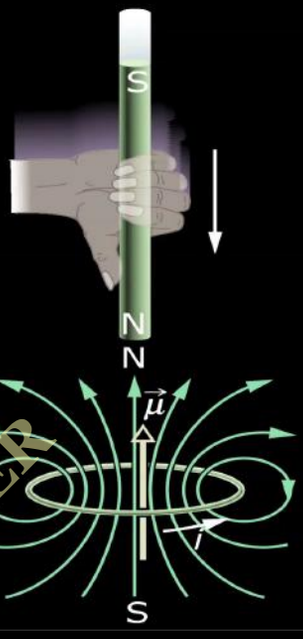
Solution: Initial flux linkage: $\phi_i = BNA$, Final flux linkage, $\phi_f = -NBA$ (Normal has rotated by 180°), thus,

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charge flown $q = \left(\frac{\Delta\phi}{R}\right) = 2NBA/R$ Ans.

LENZ'S LAW:

Earlier we learnt how to find the direction of induced EMF using Faraday's law, however there is a more convenient way of finding out the sense of the induced EMF, i.e the Lenz law.



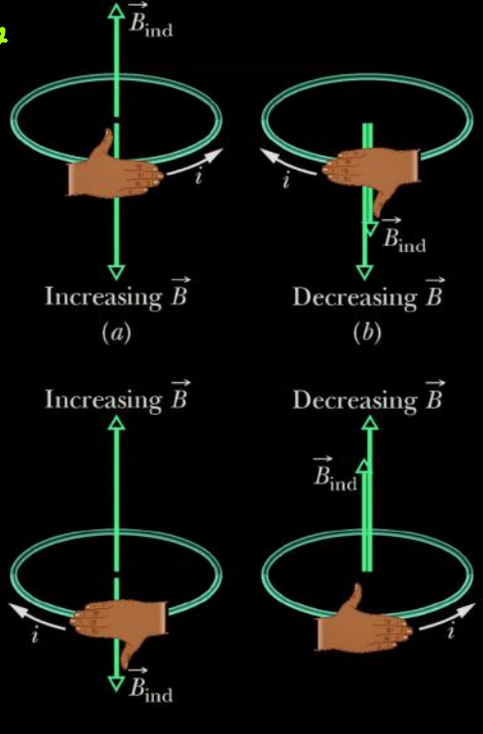
According to Lenz law, induced current (and EMF) has a direction such that the magnetic field due to the current opposes the change in magnetic flux that induces the current.

→ Simply state: Induced EMF opposes the cause of its production. (This can be thought of not only in terms of magnetic flux but also forces, torques, velocities and displacements etc). For example consider the figure above:

① Since moving the magnet nearer increases the flux in the loop (in downward direction), thus induced current increases the field in upward direction.

② Since north pole of bar magnet is coming near to cause an EMF thus induced current is produce so as to repel the magnet, thus upper side of loop becomes the North Pole of the loop.
(For finding North Pole, curl the fingers along the current and thumb gives the side of north Pole).

→ For practising Lenz's law study the adjacent examples. (Build your logic both in terms of magnetic flux as well as pole formation to cause attraction or repulsion. For instance last case (d) can be visualised as if a bar magnet's south pole were placed above the loop and bar magnet were being moved upwards, now since south pole is moving away, loop must form North pole above to attract it.

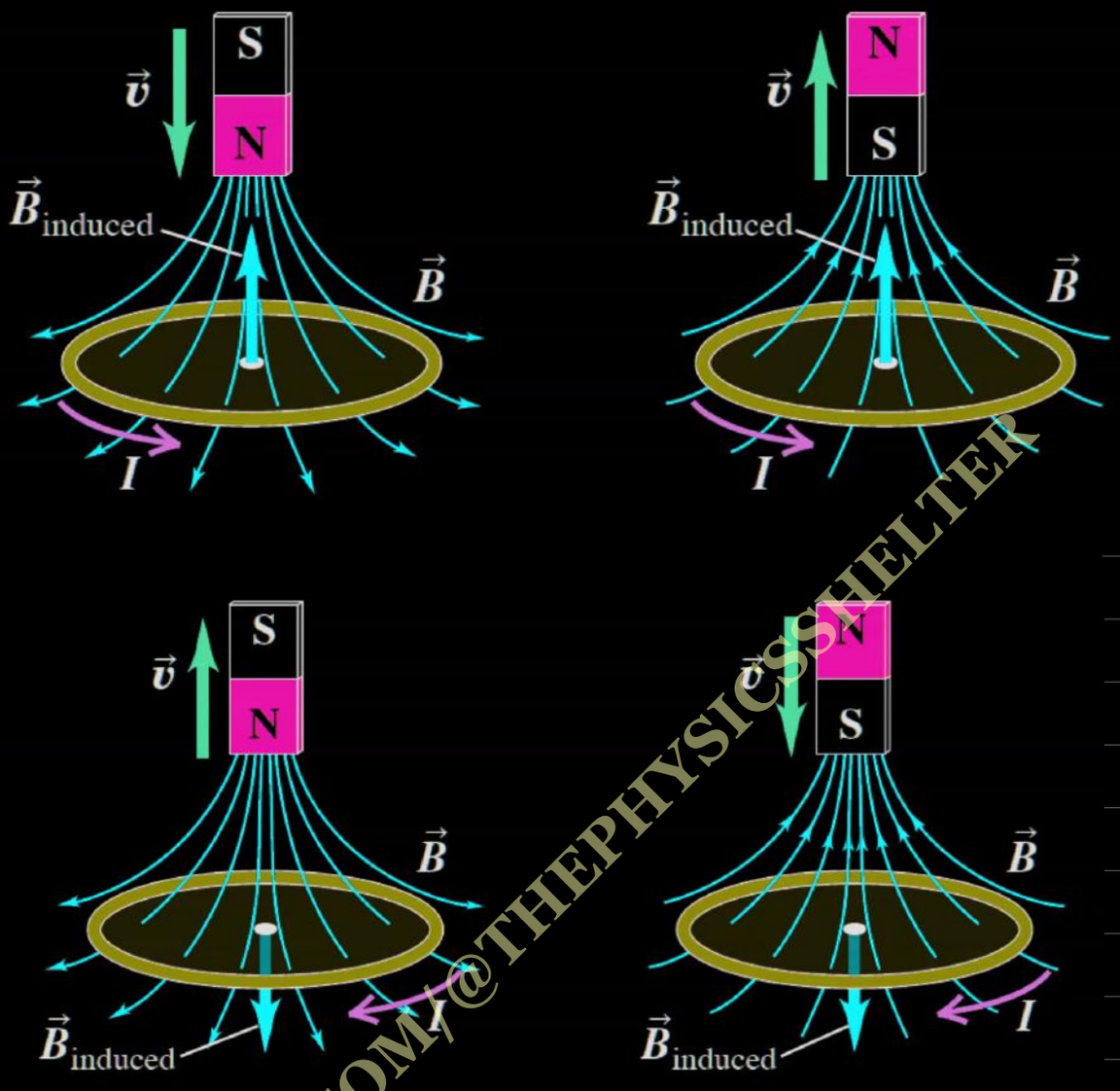


→ Lenz's law is based on energy conservation.

(Anti Lenz law will simultaneous increase magnetic and Kinetic Energy)

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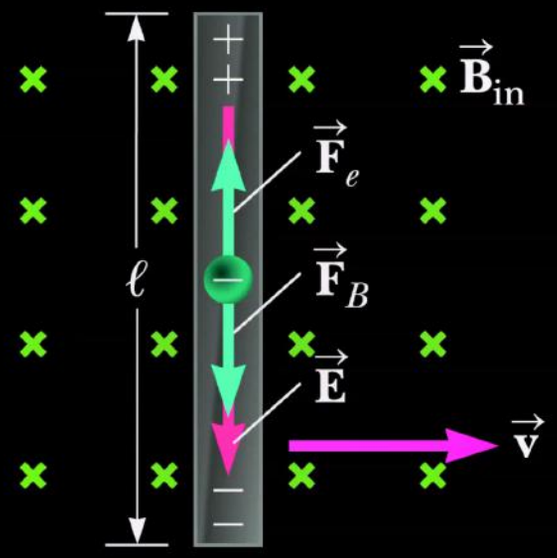
EXAMPLE: Study the following cases to find the sense of induced EMF in response to the motion of the bar magnet.



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MOTIONAL EMF:

Recall that EMF in any element of a wire is given by $d\mathcal{E} = \vec{E}^* \cdot d\vec{\ell}$ where $\vec{E}^*$ is the force of non electrostatic origin per unit charge. Now consider a wire element of elemental length vector $d\vec{\ell}$ with a velocity $\vec{v}$, due to motion of the wire charges contained in it experience a magnetic force per unit charge given as $\vec{E}^* = q\,\vec{v}\times\vec{B}/q = \vec{v}\times\vec{B}$, thus due to this force an elemental EMF developed in the element is given by $d\mathcal{E} = (\vec{v}\times\vec{B})\cdot d\vec{\ell}$

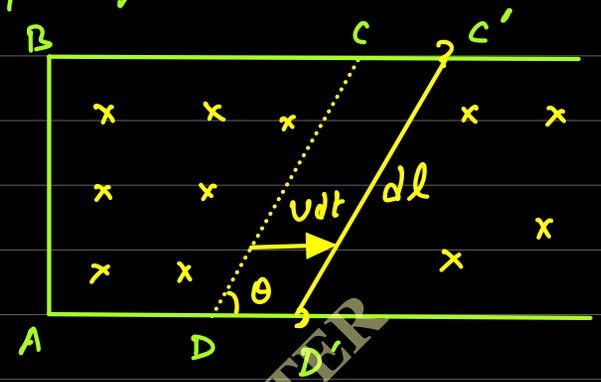


Thus total EMF in the moving wire is calculated as:

$$\boxed{\mathcal{E} = \int (\vec{v} \times \vec{B}) \cdot d\vec{l}}$$

→ Motional EMF could also be derived by faraday's law

Imagine a stationary U shaped wire frame (shown in green color) on which a wire of length dl slides from position CD to C'D' in a time dt, The increase in the area of the loop ABCD in the process is:



$dS = v dt \times dl \sin\theta$, thus change in flux is given as:

$$d\phi = B dS = B v dt\, dl \sin\theta \Rightarrow \frac{d\phi}{dt} = B v dl \sin\theta = (\vec{v} \times \vec{B}) \cdot d\vec{l}$$

→ While motional EMF problems can be solved by calculating the rate of change of flux, however Faraday's law does not transparently tell us where exactly this EMF is generate. Formulating the problem as that of motional EMF helps us visualise the mechanism of generation of this EMF precisely.

→ Since we finally have to take a dot product with $d\vec{l}$ any component of $\vec{v}$ or $\vec{B}$ parallel to the rod is useless because its cross product with any vector will give a vector whose dot product with $d\vec{l}$ equal to zero, mathematically:

$$(\vec{v} \times \vec{B}) \cdot d\vec{l} = \left[(\vec{v}_{\parallel} + \vec{v}_{\perp}) \times (\vec{B}_{\parallel} + \vec{B}_{\perp})\right] d\vec{l}$$

$$= \underbrace{(\vec{v}_{\parallel} \times \vec{B}_{\parallel}) \cdot d\vec{l}}_{=0\ (\because \vec{v}_{\parallel} \parallel \vec{B}_{\parallel})} + \underbrace{(\vec{v}_{\parallel} \times \vec{B}_{\perp}) \cdot d\vec{l}}_{=0\ \because \vec{v}_{\parallel} \times \vec{B}_{\perp} \perp d\vec{l}} + \underbrace{(\vec{v}_{\perp} \times \vec{B}_{\parallel}) \cdot d\vec{l}}_{=0\ \because \vec{v}_{\perp} \times \vec{B}_{\parallel} \perp d\vec{l}} + \underbrace{(\vec{v}_{\perp} \times \vec{B}_{\perp}) \cdot d\vec{l}}_{\text{Non Zero}}$$

(Here $\vec{v}_{\parallel}$, $\vec{v}_{\perp}$, $\vec{B}_{\parallel}$ and $\vec{B}_{\perp}$ are components parallel & $\perp$ to $d\vec{l}$).

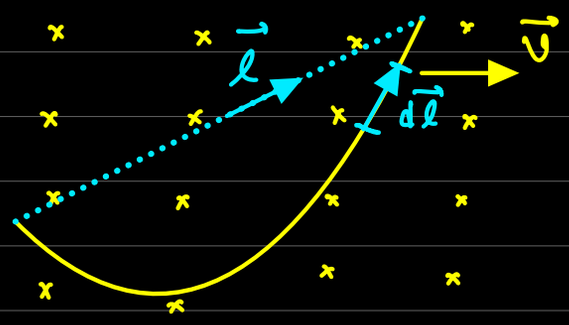
→ Thus $\boxed{d\mathcal{E} = (\vec{v}_{\perp} \times \vec{B}_{\perp}) \cdot d\vec{l}}$ This will save us a lot of calculation many times because we can drop the parallel components of $\vec{v}$ & $\vec{B}$ right away.

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EMF In A Conductor In Pure Translation In Uniform Magnetic Field:

Consider an arbitrary shaped wire moving in pure translation with a velocity $\vec{v}$ in a magnetic field $\vec{B}$, The net EMF in such a wire is given as :

$$\mathcal{E} = \int (\vec{v} \times \vec{B}) \cdot d\vec{\ell}$$, however since



$\vec{v}$ and $\vec{B}$ are constant for each $d\vec{\ell}$ so they can be taken outside the integral, thus:

$$\mathcal{E} = (\vec{v} \times \vec{B}) \cdot \int d\vec{\ell}$$, now $\int d\vec{\ell}$ is nothing

but vector joining initial position to final position thus we can write

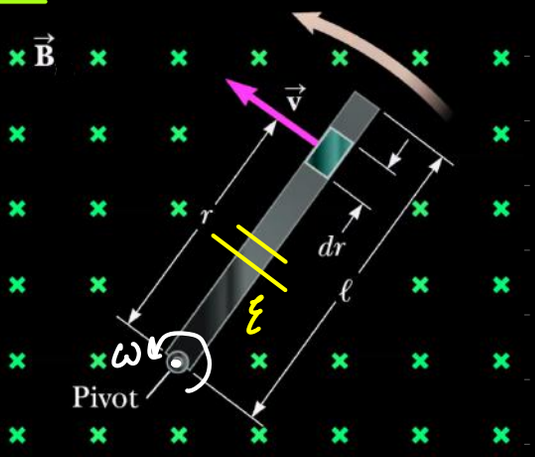
$$\boxed{\mathcal{E} = (\vec{v} \times \vec{B}) \cdot \vec{\ell}} \quad \text{or} \quad \boxed{\mathcal{E} = (\vec{v}_\perp \times \vec{B}_\perp) \cdot \vec{\ell}}$$

Here $v_\perp$ and $B_\perp$ are components of $\vec{v}$ and $\vec{B}$ perpendicular to the vector joining initial to final point of the wire.

→ Alternatively we can also consider component of $d\vec{\ell}$ along $\vec{v} \times \vec{B}$.

EMF In A Pivoted Rod Rotating In Plane ⊥ To $\vec{B}$:

Consider a rod of length ℓ pivoted about its end rotating in a magnetic field as shown. Now we consider an element of length dr at a distance r from pivot, elemental EMF in the element is given as:



$$d\mathcal{E} = [(\omega r) \times B]\, dr$$

$$\Rightarrow d\mathcal{E} = B\omega r\, dr \Rightarrow \mathcal{E} = \int_0^\ell B\omega r\, dr$$

$$\Rightarrow \boxed{\mathcal{E} = \tfrac{1}{2} B\omega \ell^2}$$

→ Polarity of the EMF can be readily found by looking at $\vec{v} \times \vec{B}$ for any element (see figure)

→ Even if there were some component of B in plane of rotation, it will have no effect on EMF because its cross product with any

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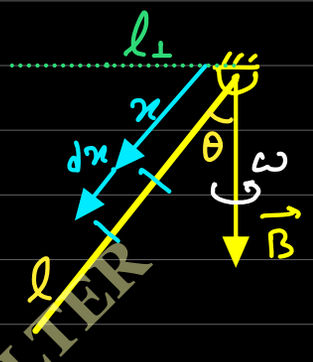
velocity in the plane will lead to a vector ⊥ to plane of rotation and hence yield zero upon doing dot product with $d\vec{l}$. Thus for such rods we can write $\mathcal{E} = \frac{1}{2} B_\perp \omega l^2$ where $B_\perp$ is component of B perpendicular to plane of rotation.

Rod Executing Conical Pendulum With $\vec{B} \parallel \vec{\omega}$

Consider a rod rotating as a conical pendulum, Here component of B parallel to rod contributes nothing to EMF. Thus elemental EMF:

$$d\mathcal{E} = (\omega x \sin\theta)\, B \sin\theta\, dx = B\omega \sin^2\theta\, x\, dx$$

Thus $\mathcal{E} = \int d\mathcal{E} = B\omega \sin^2\theta \int x\, dx = \frac{1}{2} B\omega (l\sin\theta)^2$



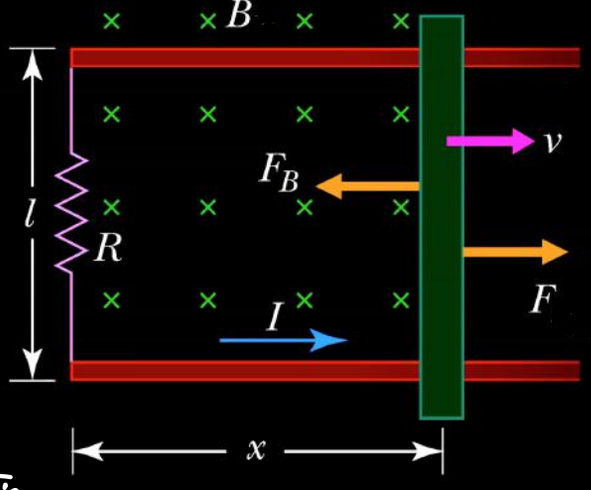
Now note that $l\sin\theta$ is component of l ⊥ to B thus:

$$\boxed{\mathcal{E} = \frac{1}{2} B\omega l_\perp^2}$$

→ Think & Prove at home: For rod executing conical pendulum with $\vec{\omega} \perp \vec{B}$ the EMF is sinusoidal with amplitude $\frac{1}{2} B\omega \sin\theta\cos\theta\, l^2$ (Time average value is zero).

Slider Rail System With Resistance:

This is a category of problems where we have a slider being pulled on rails and some circuit elements in the circuit being completed through the slider (A resistor, capacitor, inductor or a combination thereof). Such questions typically combine F=ma equations along with Induced EMF and magnetic force ideas. Net rate of work done by the pulling force is equal to sum of rate of kinetic energy, rate of energy storage and energy dissipation. In our problem at hand, a slider of mass m is being pulled by a constant force F over perfectly conducting frictionless rails and we wish to find out velocity as a function of time along with a few other things in the circuit. Let us now frame equations:



$$F - IlB = ma \quad (1)$$

$$I = Blv/R \quad (2)$$

Now using (2) in (1) $F - B^2l^2v/R = m\,dv/dt$ (3) thus we

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have a differential equation in v and t which we can easily integrate to find $v(t)$. (Student is advised to calculate $v(t)$ if $v(0)=0$, the result we get: $v = \frac{FR}{B^2l^2}\left(1 - e^{-B^2l^2t/mR}\right)$ ④

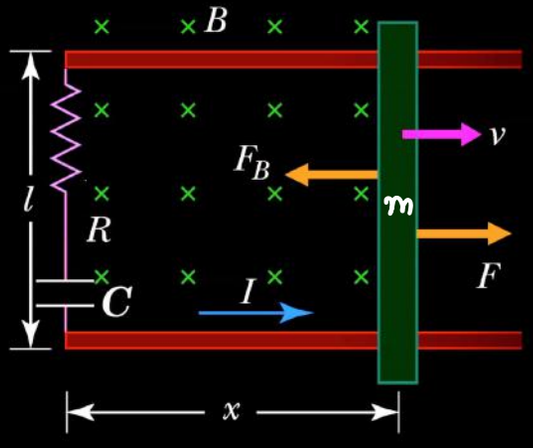
→ It can be seen that it is a case of saturating velocity with terminal velocity being FR/B^2l^2. We could also obtain this terminal velocity by direct force balance i.e $F = \left(\frac{Blv}{R}\right)lB$ ($\because F = F_B$)

→ Multiplying ③ by v and rearranging we get:

$$Fv = mv\frac{dv}{dt} + \left(\frac{Blv}{R}\right)^2 R \implies \vec{F}\cdot\vec{v} = \frac{d}{dt}\left(\frac{1}{2}mv^2\right) + I^2R$$

This simply first law of thermodynamics i.e Power developed by external agent is equal to rate of heat dissipation and rate of increase of kinetic energy of the system.

EXAMPLE: Consider adjacent figure. Mass of slider is m, rails are perfectly conducting and frictionless (a) develop a differential equation to solve for $i(t)$ (b) find the solution and analyse its nature.



Solution:

$$F - IlB = m\dot{v} \quad ①$$

$$Blv = IR + q/C \quad ②$$

Now from ②, $Bl\dot{v} = R\dot{I} + I/C$ ③, Using ① in ③

$$\left(\frac{F - IlB}{m}\right)Bl = R\dot{I} + I/C \quad ④$$

Now ④ is a variable separable differential equation, with $I=0$ @ $t=0$

Integrating we get:

$$I(t) = \frac{FBCl}{m + B^2l^2C}\left[1 - e^{-(m+B^2l^2C)t/mRC}\right]$$

→ It can be seen at $t=\infty$, current becomes constant and equal to $\frac{FBCl}{m+B^2l^2C}$

however $F - BIl \neq 0$ thus slider will start moving with a constant acceleration $a = (F - BIl)/m = F/(m + B^2l^2C)$

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→ If resistor were not there we can obtain the solution by thinking the limit $R \to 0$, in this case, the current becomes constant immediately and slider moves with constant acceleration $F/(m + B^2\ell^2 C)$

→ Had we already known that final state will have a constant acceleration, we could directly obtain steady state acceleration directly from ① & ③

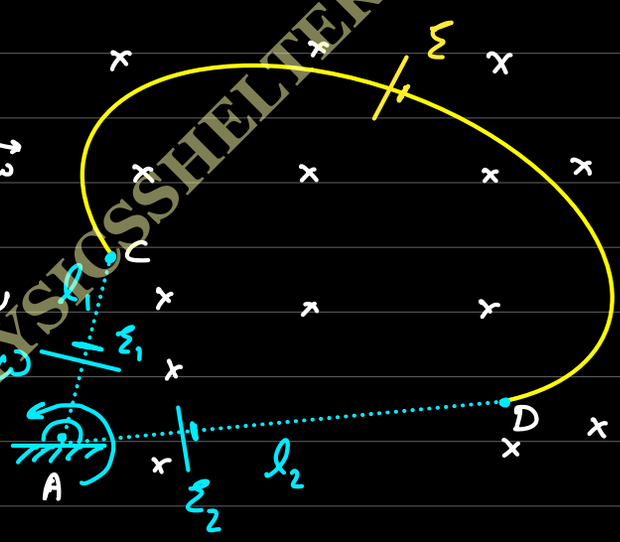
Thus: $F - I\ell B = ma$ ① and $B\ell a = I/C$ ③' $(\because \dot{I} = 0)$

From ① & ③':

$$F - B^2\ell^2 C a = ma \implies a = F/(m + B^2\ell^2 C).$$

EMF In Rotating Arbitrary Wire ($\vec{B} \perp$ To Plane Of Rotation):

Consider an arbitrary shaped wire (yellow) rotating about its instantaneous axis A while maintaining it shape. $\vec{B}$ is $\parallel$ to $\vec{\omega}$ and the plane of the loop is $\perp$ to $\vec{B}$. Let its end points be located at distances ℓ_1 and ℓ_2 from A as shown, now we complete a closed loop by joining the ends of the wire by dotted lines.



There is no flux change in the loop thus created since area of loop is constant and perpendicular to uniform $\vec{B}$. Now let EMF in the wire be Σ and that in ℓ_1 and ℓ_2 portions be ε_1 and ε_2 as shown. Now net EMF along the loop:

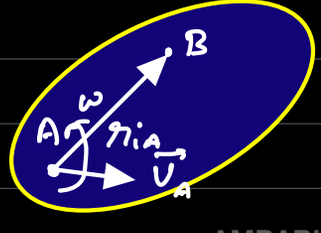
$$\varepsilon_2 - \Sigma - \varepsilon_1 = 0 \implies \Sigma = \varepsilon_2 - \varepsilon_1 = \frac{1}{2} B\omega(\ell_2^2 - \ell_1^2)$$

Thus $$\boxed{|\Sigma| = \frac{1}{2} B\omega |(\ell_2^2 - \ell_1^2)|}$$

→ For calculating the polarity of Σ always visualise by actually completing the loop.

EMF In An Arbitrary Rigid Conductor In General Motion:

Recall that velocity of any particle in a rigid body is given as $\vec{v}_i = \vec{v}_A + \vec{\omega} \times \vec{r}_{i,A}$



Thus EMF in any segment of length $d\vec{\ell}$ near i^{th} particle is simply $[(\vec{v}_A + \vec{\omega} \times \vec{r}_{i,A}) \times \vec{B}] \cdot d\vec{\ell}$

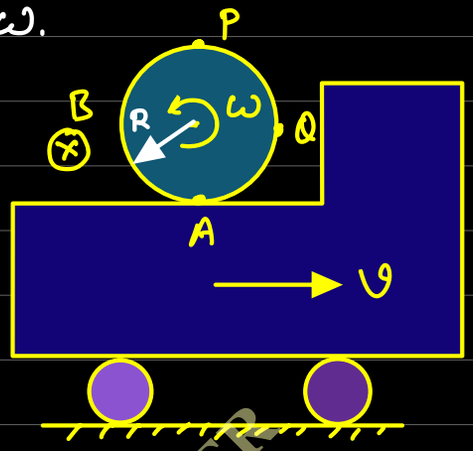
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By distributive law $d\mathcal{E} = (\vec{v}_A \times \vec{B}) \cdot d\vec{\ell} + [(\vec{\omega} \times \vec{r}_{i:A}) \times \vec{B}] \cdot d\vec{\ell}$

Thus EMF can be found by superposition of two terms:

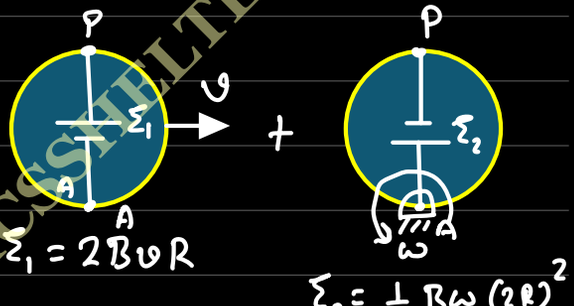
① Translating the entire body with v_A

② Rotating the entire body about A with ω.



EXAMPLE: An aluminium cylinder is rolling on a surface of a moving truck with magnetic field in to the page. Find EMF induced between point (a) A and P (b) A and Q (c) Q and P.

Solution: (a) We can model the problem in to superposition of translation of A and rotation about A (see adjacent figure).

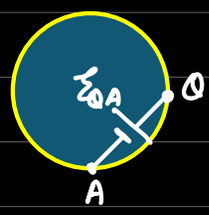


Thus: $\mathcal{E}_{PA} = \mathcal{E}_1 - \mathcal{E}_2 = 2Bv R - 2B\omega R^2$. Ans. ↑

(b) By a similar logic as part A, $\mathcal{E}_{QA}$ can be obtained by superposition of translation of point A and rotation about point A. Thus $\mathcal{E}_{QA} = Bv R - \frac{1}{2} B\omega (\sqrt{2}R)^2$

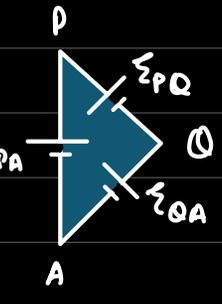


Thus $\mathcal{E}_{QA} = Bv R - B\omega R^2$ Ans



(c) Now imagine closed triangle AQP. We know that net flux through this loop remains constant, thus along the loop $\mathcal{E}_{PQ} + \mathcal{E}_{QA} - \mathcal{E}_{PA} = 0$

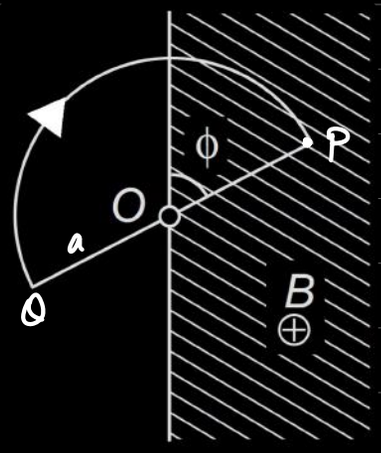
$\Rightarrow \mathcal{E}_{PQ} = \mathcal{E}_{PA} - \mathcal{E}_{QA} = Bv R - B\omega R^2$ Ans.



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Example : Irodov 3.292:

A wire loop enclosing a semi-circle of radius a is located on the boundary of a uniform magnetic field of induction B (See Fig.). At the moment $t = 0$ the loop is set into rotation with a constant angular acceleration α about an axis O coinciding with a line of vector $\vec{B}$ on the boundary. Find the emf induced in the loop as a function of time t. Draw the approximate plot of this function. The arrow in the figure shows the emf direction taken to be positive.

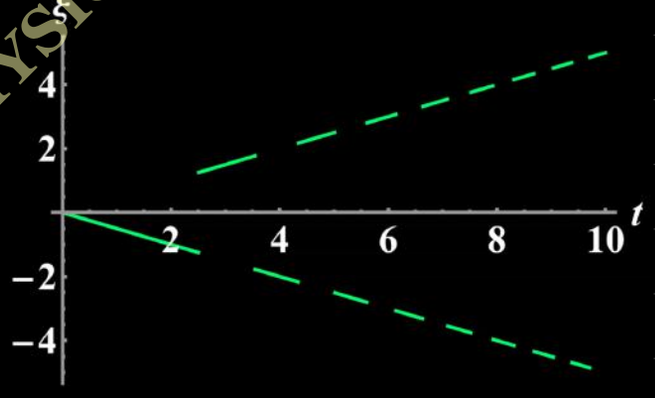


Solution: If ϕ is between 0 and 180°, $EMF = -\frac{1}{2} B\omega a^2$ (Against the arrow therefore -ve) while if ϕ is between 180° and 360°, EMF is $+\frac{1}{2} B\omega a^2$ and this pattern repeats alternately. Thus if θ is the angle turned, EMF is given as:

$\mathcal{E} = -\frac{1}{2} B\omega a^2 (-1)^{[\theta/\pi]}$, where $[\theta/\pi]$ represents greatest integer function.

Also $\omega = \alpha t$ and $\theta = \frac{1}{2}\alpha t^2$. For a qualitative idea, adjacent figure shows a qualitative plot of EMF for $Ba^2 = 1$ and $\alpha = 1\ rad/s^2$.

Note that dashes keep becoming smaller and smaller due to increasing ω and hence more frequent change in polarity. Also magnitude of EMF keep increasing.

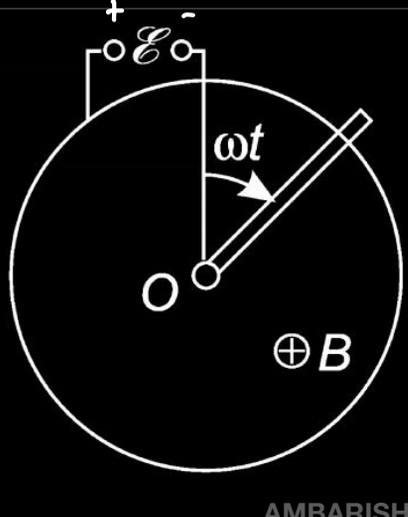


→ Alternatively the problem could also be formulated using $d\phi/dt$ by expressing flux in enclosed area as a function of time.

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Example : Irodov : 3.295:

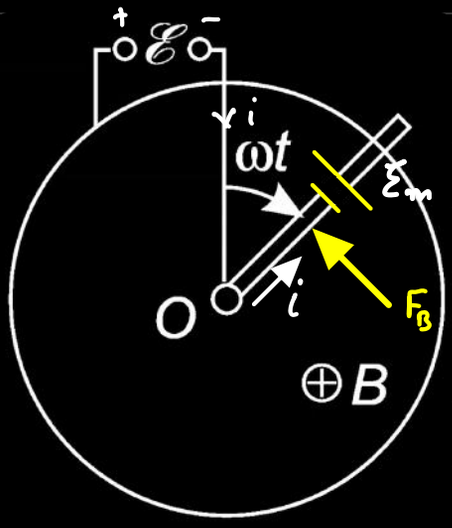
A metal rod of mass m can rotate about a horizontal axis O, sliding along a circular conductor of radius a (See Fig.). The arrangement is located in a uniform magnetic field of induction B directed perpendicular to the ring plane. The axis and the ring are connected to an emf source to form a circuit of resistance R. Neglecting the friction, circuit inductance, and ring resistance, find the law according to which the source emf must vary to make the rod rotate with a constant angular velocity ω.



Solution: Motional EMF in the loop $\mathcal{E}_m = \frac{1}{2} B\omega a^2$.

Thus current in the loop:

$$i = (\mathcal{E}_m - \mathcal{E})/R$$



Now for rod to rotate with constant ω, magnetic torque must be balancing the torque of gravity. Magnetic force is $\perp$ to the rod and can be applied at mid point due to uniform distribution. Thus:

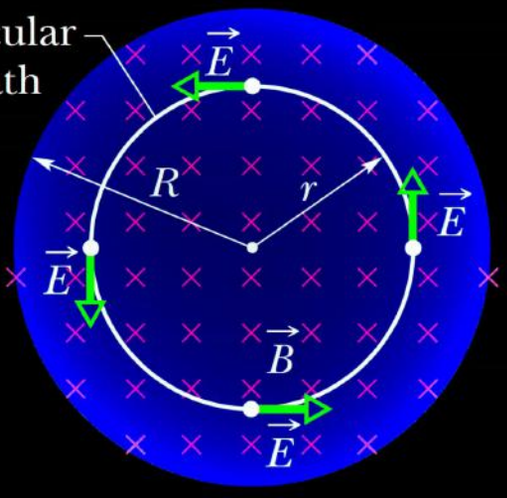
$$i\,a\,B \times \frac{a}{2} = mg \sin\omega t \times (a/2)$$

Thus $\frac{1}{R} \cdot \left(\frac{1}{2} B\omega a^2 - \mathcal{E}\right) a B = mg \sin\omega t$,

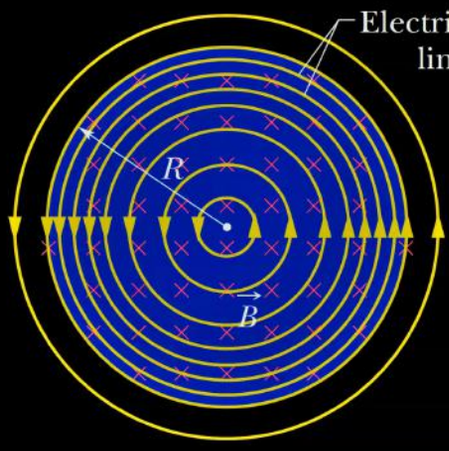
Thus $\mathcal{E} = \frac{B\omega a^2}{2} - \frac{mg \sin\omega t R}{aB}$ Ans. [There is a sign mistake in Irodov Answer].

Induced Electric Field (Non Electrostatic Electric Field):

Consider a cylindrical region carrying a time varying magnetic field in to the page. Let $\dot{B}$ be the rate of change of field assumed uniform. R is the radius of cylindrical region. Now consider a loop of radius r ($r < R$) in the region, in this loop, $\phi = B\pi r^2 \Rightarrow d\phi/dt = \dot{B}\pi r^2$



This is magnitude of induced EMF which is nothing but integral of a non conservative pushing force. This means there must be a pushing force/unit charge E^* created in the region, such that $\mathcal{E} = \int \vec{E}^* \cdot d\vec{\ell}$.



This E^* is called non conservative Electric field. We can find E^* as follows:

For $r < R$, $E^* \times 2\pi r = \mathcal{E} = \dot{B} \times \pi r^2$

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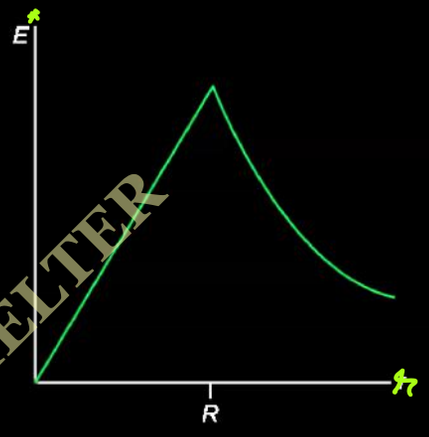
$\Rightarrow E^* = \dot{B}r/2$, this increases linearly with r for $r < R$.

For $r > R$:

$$E^* \times 2\pi r = d/dt\,[B \times \pi R^2] = \dot{B}\pi R^2$$

$\Rightarrow E^* = \dot{B}R^2/2r$. Thus we have formulae:

$\boxed{E^* = \dot{B}r/2}$ for $r < R$ and $\boxed{E^* = \dot{B}R^2/2r}$ For $r > R$



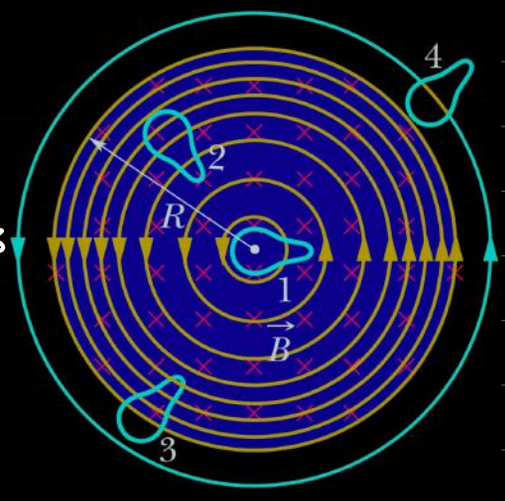
→ E^* field lines are tangential

→ Field is largest near the periphery of the cylindrical region and then declines hyperbolically.
The adjacent graph shows variation of E^* vs r.

→ The E^* field is of the form $\frac{C}{r}\hat{\theta}$,

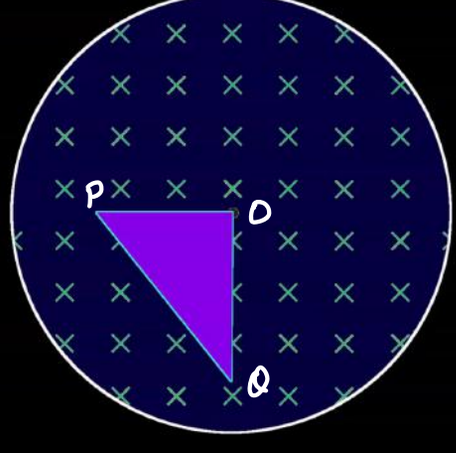
for the region outside cylinder, recall this form of field has $\int E^* \cdot dl = 0$ (special behavior like a conservative force) so long as a loop is completely in the region outside the magnetic field and does not go around the centre full turn.

→ Thus in adjacent figure, $\int \vec{E}^* \cdot d\vec{l}$ along loops 1, 2, 3 and 5 is non-zero, while for loop 4 it is zero.



→ EMF along any radial line from the centre of the cylinder is zero because everywhere E^* is perpendicular to radius thus $\vec{E}^* \cdot d\vec{l}$ will be zero along radius.

→ To find EMF along any arbitrary path PQ can be found by joining its extremities to the centre to form a closed area, now EMF in this path is simply $\dot{B} \times A$ where A is the enclosed area. (This is because there is no EMF in radial lines, so entire EMF of the loop is same as that along PQ.



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→ If an arbitrary curve is going around the centre several times calculate the area very carefully (Try to find area of each 360° turn piecewise and add up all such areas, i.e add up the EMF of all the turns with proper sign).

EXAMPLE (ORIGINAL):

A spiral path is created in a region of magnetic field with polar equation of spiral given as $r = a\theta$ where a is some constant. Find EMF induced in the spiral from centre to a general r ($r < R$).



Solution: Area swept by a line from centre of the field $dA = \frac{1}{2} r^2 d\theta$, thus total area swept till a general r, $A = \int \frac{1}{2} r^2 d\theta$

$$\Rightarrow \int_0^r \frac{1}{2} r^2 \frac{1}{a} dr = (r^3/6a) \quad \text{thus } EMF = \dot{B}A = \dot{B}r^3/6a \text{ Ans.}$$

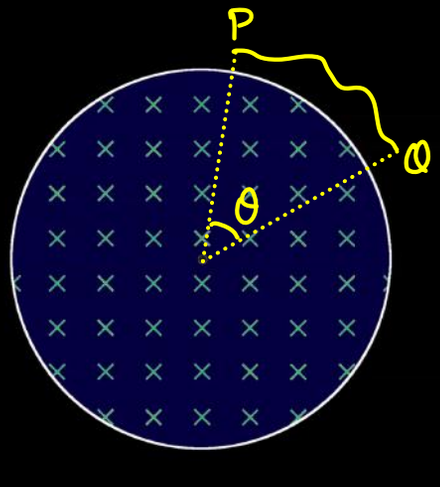
EXAMPLE: For situation in previous problem, what is EMF induced from R to r ($r > R$).

Solution: For outside point, $E^* = \dot{B}R^2/2r$, Now $|\vec{E}^* \cdot \vec{dR}| = \frac{\dot{B}R^2}{2r} \times r\,d\theta$

$$\Rightarrow d\mathcal{E} = \frac{\dot{B}R^2}{2} d\theta = \frac{\dot{B}R^2}{2} \cdot \frac{1}{a} dr,$$

Thus EMF induced from R to r, $\mathcal{E} = \frac{\dot{B}R^2}{2a} \int_R^r d\theta = \frac{\dot{B}R^2}{2a}(r - R)$ Ans.

→ For any arbitrary curve outside the time varying magnetic field, EMF only depends on angle subtended by extremities at the centre (Flux is associated only with a perfect sector). Thus $\mathcal{E} = \dot{B} \times \frac{\theta}{2\pi} \times \pi R^2$ or $\mathcal{E} = \dot{B}\theta R^2/2$

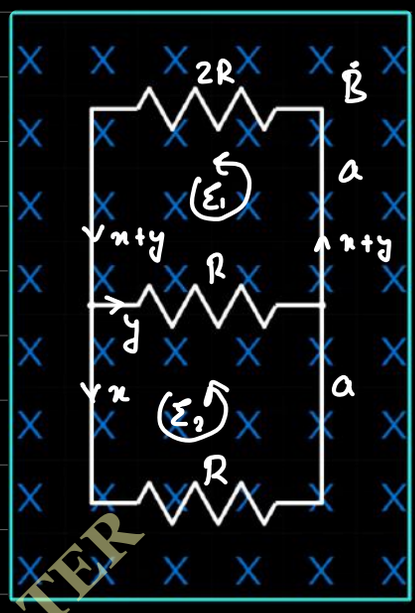


→ θ will be more than 2π if PQ goes around the circle more than once.

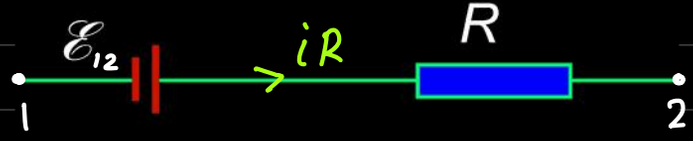
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Modified Kirchoff's Laws In Time Varying Fields:

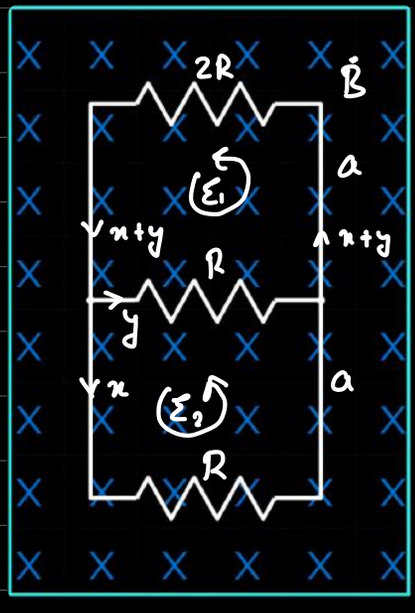
Consider the circuit shown in the adjacent figure, the circuit is immersed in a time varying magnetic field $\dot{B}$ in to the page. The circuit comprises of two square loops with resistances as shown. Here we do not know localized EMF in any branch, however we know net EMFs in the loops i.e $\mathcal{E}_1 = \mathcal{E}_2 = a^2\dot{B}$. Thus to apply Kirchoff's laws, we can't consider EMFs branchwise. In such situations however we can still write Kirchoff's laws for complete loops as: $\Sigma iR = \mathcal{E}$ where ΣiR represents current and resistance products along any loop. Infact here EMFs are distributed in space and unlike case of cylindrical magnetic field, here shape of induced field lines is not precisely known, that's why we don't know branchwise EMFs.



→ Recall from current electricity that $V_1 - V_2 = -\mathcal{E}_{12} + iR$, thus without knowing EMF, it is not possible to find out potential differences. Thus with non-cylindrical fields, it will be generally not possible to find potential differences between points unless due to some kind of symmetry arguments we precisely know the EMFs in various branches.



Example: In the shown adjacent circuit find the currents x and y if B is uniform and changing at a rate $\dot{B}$ in to the page.



Solution: For upper loop: $(x+y)\times 2R + yR = \dot{B}a^2$ ①

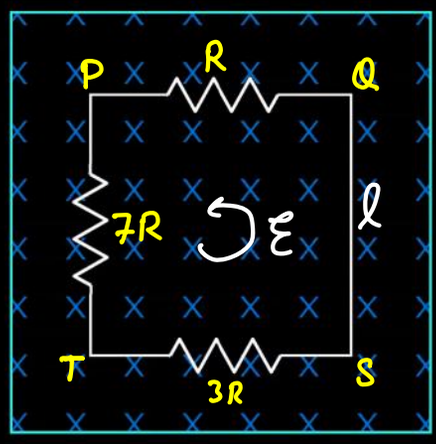
For lower loop: $xR - yR = \dot{B}a^2$ ②

Solving ① & ②

$$x = \frac{4}{5}\frac{\dot{B}a^2}{R}, \quad y = -\frac{1}{5}\frac{\dot{B}a^2}{R} \text{ etc.}$$

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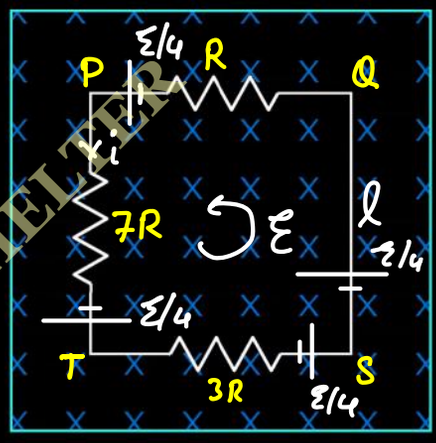
Example: Consider adjacent figure. A square loop of side ℓ is placed perfectly symmetrically in a square region of uniform magnetic field increasing inward at a rate $\dot{B}$. Find:
(a) Current in the loop (b) $V_P - V_Q$ (c) $V_S - V_Q$



Solution: (a) $i = \mathcal{E}/R_{eff} = \dfrac{\dot{B}\ell^2}{11R}$ Ans.

(b) Since there is perfect symmetry therefore, here we can find the EMF each branch as 1/4 of the EMF in entire loop. Now considering branch QP: $V_Q - V_P = iR - \mathcal{E}/4$

$$\Rightarrow V_P - V_Q = \mathcal{E}/4 - iR = \frac{\dot{B}\ell^2}{4} - \frac{\dot{B}\ell^2}{11} = \frac{7}{44}\dot{B}\ell^2 \text{ Ans.}$$



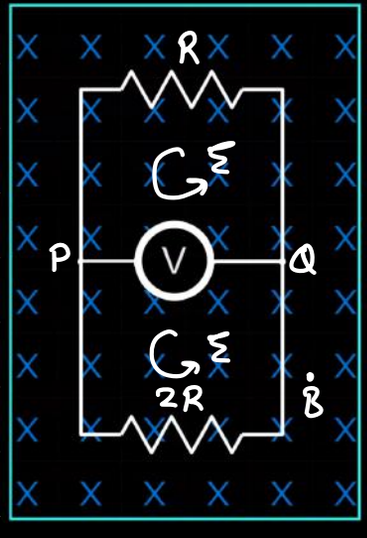
(c) Considering branch SQ: $V_S - V_Q = -\mathcal{E}/4 = -\dfrac{\dot{B}\ell^2}{4}$ Ans.

→ Recall that potential difference is nothing but negative integral of conservative electric field i.e. $V_Q - V_P = -\int_P^Q \vec{E}\cdot d\vec{\ell}$. This conservative field appears due to appearance of static charges in the conductors to push same current through all resistors despite their different resistances.

→ Had there been no symmetry in the problem (say due to square being placed off-centre) there would be no easy manual method to solve this problem.

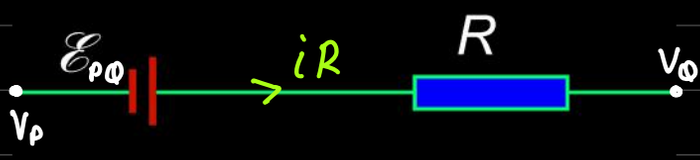
What Does A Voltmeter Read In Time Varying Fields?

Recall the mechanism of a moving coil Galvanometer, the deflection in coil is proportional to the current flowing through it so the voltmeter having this mechanism measures the value of iR where i is the current through voltmeter coil and R is the coil resistance. A common misconception is that the voltmeter measures the potential difference, However this is not correct in time varying fields.



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For instance in the figure shown above, $V_P - V_Q = -\mathcal{E}_{PQ} + iR$. This issue is exclusive to time varying fields because there is an induced EMF in the wires connecting the voltmeter to the circuit, which was not the case in the circuits that we came across in the current electricity chapters.



→ Student should very clearly understand the difference between following three quantities:

① Voltmeter reading (This is simply iR)

② Potential Drop (This is simply $iR \pm \mathcal{E}$ depending on direction)

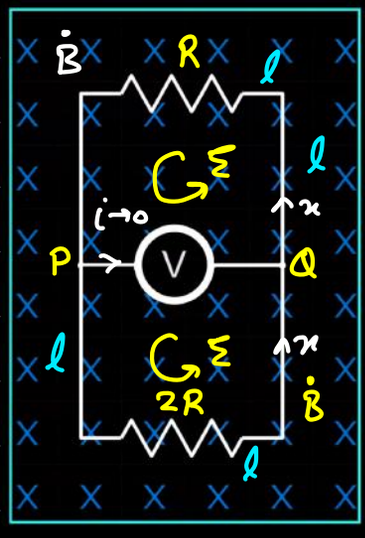
③ EMF (This is simply $\mathcal{E}$).

→ Potential drop is an effect of conservative field while EMF is an effect of a non-conservative field.

→ For an ideal voltmeter, $i = 0$ and $R \to \infty$ however the product iR can still be finite. The readings of voltmeter can be obtained by applying loop laws to the loops containing voltmeters just as we did for finding currents in a previous example.

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EXAMPLE: In the circuit shown in adjacent figure, (a) Find the voltmeter reading (Both loops are $\ell \times \ell$ squares) (b) Assuming asymmetric placement of circuit in time varying field, is it possible to find EMF and potential difference in branch PQ? (c) If voltmeter were shifted a distance $\ell/2$ upwards in the figure, what will be the voltmeter reading? (Voltmeter is ideal, $\dot{B}$ is known).



Solution: (a) EMF in both loops is $\dot{B}\ell^2$. Now applying loop law to complete rectangular loop, current in the loop is $x = 2\mathcal{E}/3R$.

Now applying loop law to upper loop: (R_v is voltmeter resistance).

$$iR_v + xR = \dot{B}\ell^2 \Rightarrow iR_v = \dot{B}\ell^2 - xR = \dot{B}\ell^2 - \frac{2}{3}\dot{B}\ell^2$$

$$\Rightarrow iR_v = \frac{1}{3}\dot{B}\ell^2$$, Thus voltmeter reading is $\dot{B}\ell^2/3$ ohm.

(b) To know potential difference exact EMF in the branch is required, with asymmetric placement, EMF can't be known and hence

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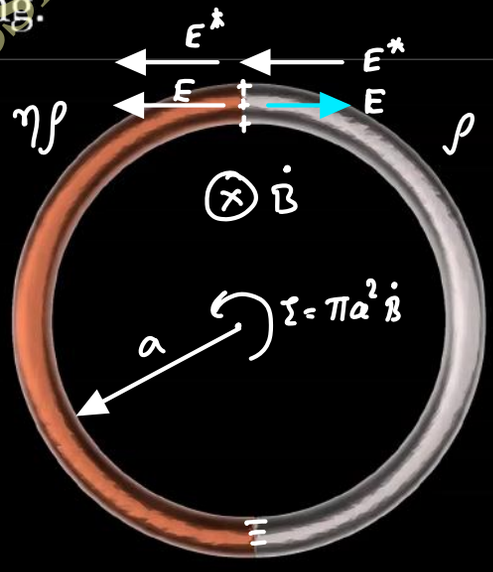
potential difference can't be known.

(c) If now voltmeter were shifted by a distance $\ell/2$ upwards, EMF of upper loop becomes $\dot{B}\ell^2/2$. Thus applying loop law to upper loop, $iR_v + nR = \dot{B}\ell^2/2 \Rightarrow iR_v = \frac{\dot{B}\ell^2}{2} - nR = \dot{B}\ell^2\left(\frac{1}{2} - \frac{2}{3}\right)$

$\Rightarrow iR_v = -\dot{B}\ell^2/6$, which means current will be from right to left in the voltmeter branch.

EXAMPLE: IRODOV 3.310:

A long solenoid of cross-sectional radius a has a thin insulated wire ring tightly put on its winding; one half of the ring has the resistance η times that of the other half. The magnetic induction produced by the solenoid varies with time as $B = bt$, where b is a constant. Find the magnitude of the electric field strength in the ring.



Solution: Let resistivity of two parts of the ring be ρ and $\eta\rho$. Now to push equal currents there will be accumulation of charges such that:

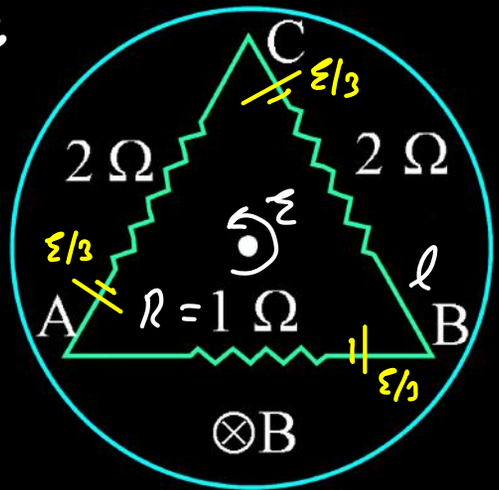
$$\frac{E + E^*}{\eta\rho} = \frac{E^* - E}{\rho}$$

$$\Rightarrow \frac{E^* - E}{E^* + E} = \frac{1}{\eta} \Rightarrow \frac{\eta - 1}{\eta + 1} = \frac{E}{E^*}$$

Now $E^* = \dot{B} \times \pi a^2/(2\pi a) = ab/2$

$E = \left(\frac{\eta - 1}{\eta + 1}\right)\frac{ab}{2}$ Ans. [Question intends to ask only conservative field].

EXAMPLE: In the shown figure, each side of triangle is $\ell = 2$m and $\dot{B} = \sqrt{3}$ T/sec inwards. Find $V_A - V_B$ assuming the triangle is placed with centroid coinciding with centre of circle.



Solution: Loop EMF $\Sigma = \dot{B} \times \frac{1}{2}\ell \times \ell \sin 30°$

$\Rightarrow \Sigma = \frac{\sqrt{3}}{4}\ell^2\dot{B} = 3$ Volts. Due to symmetry, this is equally distributed in three sides.

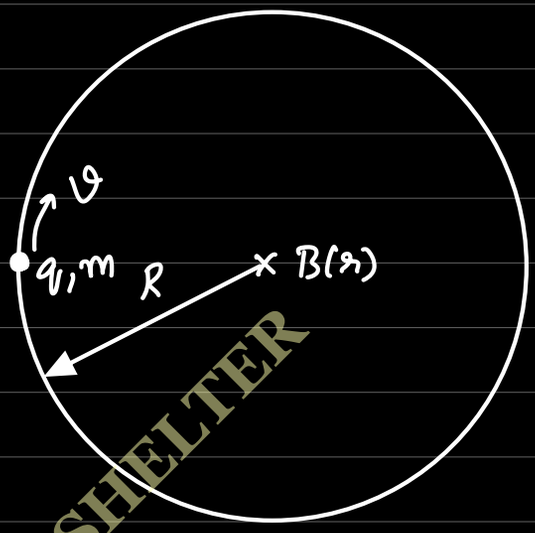
Now along branch A-D, $iR - \Sigma/3 = V_A - V_B$

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also $i = \mathcal{E}/5R \Rightarrow V_A - V_B = \mathcal{E}/5 - \mathcal{E}/3 = -2\mathcal{E}/15 = -2/5$ Volt.

→ Even if the triangle were asymmetrically placed in cylindrical field, we could still find EMF precisely in each side because shape of E^* is precisely known.

BETATRON: Recall that a cyclotron is unsuitable for accelerating electrons due to relativistic changes in mass of electron. A betatron takes care of this problem by rotating the charged particles along a circle (i.e constant radius) in a region of time varying cylindrically symmetric field. Let B_{avg} be the space average of magnetic field in the cylindrical region. Thus $\phi = B_{avg} \times \pi R^2$, now if we have a time varying magnetic field, $E^* = \frac{\dot{\phi}}{2\pi R} = \dot{B}_{avg} R/2$ i.e $E^* = \dot{B}_{avg} R/2$ ①



Also, $mv^2/R = qvB_p$ or $R = mv/qB_p$ ② where B_p is field at the periphery i.e at radius R.

Thus tangential acceleration: $\dot{v} = \frac{qR}{m}\dot{B}_p = \frac{E^* q}{m}$ ③

Now using ① in ③ : $R\dot{B}_p = \dot{B}_{avg} R/2 \Rightarrow \boxed{\dot{B}_p = \dot{B}_{avg}/2}$ ④

Further if initial magnetic field is zero then, integrating ④, we get

$\boxed{B_p = B_{avg}/2}$ ← BETATRON CONDITION.

This condition is known as Betatron condition. There can be infinitely many cylindrically symmetric functions that can satisfy this condition. Two commonly seen fields in problems satisfying this condition are:

① Region between tightly fitting nearly identical solenoids ($B_{avg} = 2\mu_0 ni$, $B_p = \mu_0 ni$ (∵ Recall the field at outside point is zero due to infinite solenoid).

② $B = B_p R/r$. $\left(B_{avg} = \frac{1}{\pi R^2}\int_0^R (B_p R/r) \times 2\pi r\, dr = 2B_p \right)$.

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→ Energy gained per revolution is $\Sigma q = \dot{B}_{avg} \pi R^2 q$

→ In case of very high velocities, relativistic effects have to be considered. Following formulae can be used:

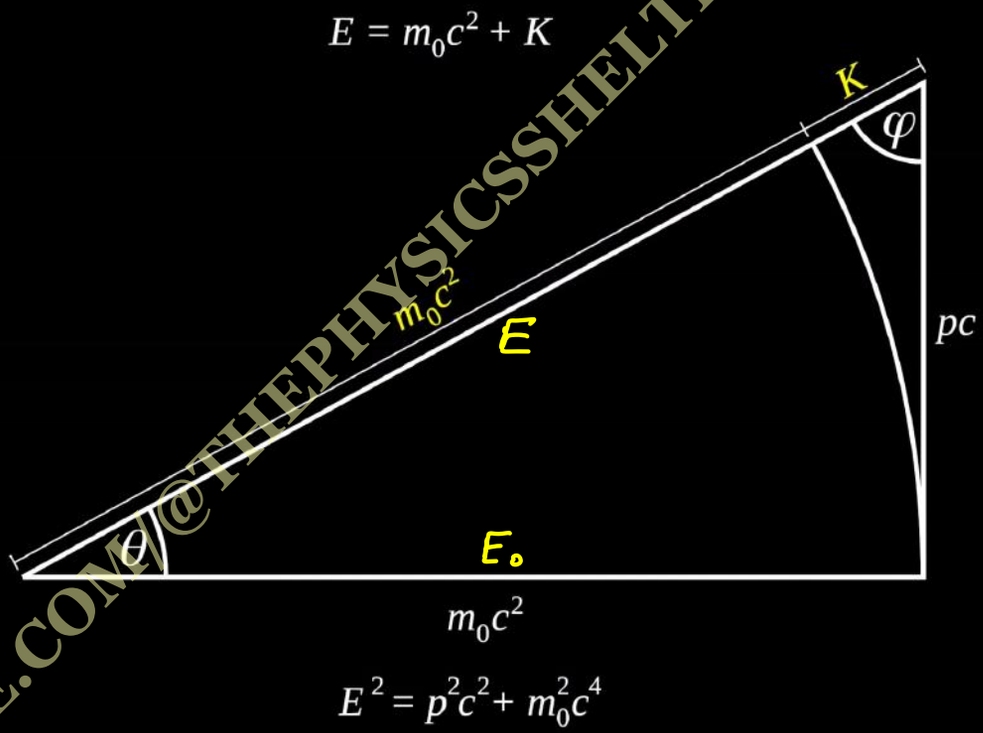
① $m = m_0 / \sqrt{1 - v^2/c^2}$

② $KE = mc^2 - m_0 c^2$ where m_0 is rest mass.

③ For electron $m_0 = 511\ keV/c^2$ if Electron energy is significantly greater than 511 keV then all of it may be considered as KE

④ For relativistic momentum and Energies the following triangle can be helpful (Used in two problems in NCERT but beyond JEE Syllabus):

E: Total Energy
m_0: Rest mass
m: Mass $= m_0/\sqrt{1 - v^2/c^2}$
p: Momentum $= mv$
K: Kinetic Energy.
E_0: Rest mass energy $= m_0 c^2$



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EXAMPLE: IRODOV 3.402:

In a betatron the magnetic flux across an equilibrium orbit of radius $r = 25$ cm grows during the acceleration time at practically constant rate $\dot{\Phi} = 5.0$ Wb/s. In the process, the electrons acquire an energy $W = 25$ MeV. Find the number of revolutions made by the electron during the acceleration time and the corresponding distance covered by it.

Solution: 25 MeV >> 511 keV thus entire 25 MeV may be considered as Kinetic Energy (i.e Gained as a result of acceleration).

Now let N be the number of revolutions, thus $N\dot{\Phi}q = W \Rightarrow N = W/q\dot{\Phi}$ Now $(q = e)$, putting Values, $W/e\dot{\Phi} = 5 \times 10^6$ revolutions, $S = N \times 2\pi r = 8 \times 10^2$ km.

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Example: Irodov 3.404:

Using the betatron condition, find the radius of a round orbit of an electron if the magnetic induction is known as a function of distance r from the axis of the field. Examine this problem for the specific case $B = B_0 - ar^2$, where B_0 and a are positive constants.

Solution: Let orbit radius be R, now $B_p = 1/2\ B_{avg}$, thus:

$$B_0 - aR^2 = \left(\frac{1}{\pi R^2}\int_0^R (B_0 - ar^2) \times 2\pi r\, dr\right) \times \frac{1}{2} \quad (1)$$

Solving (1), we get $R = \sqrt{2B_0/3a}$ Ans.

Example Irodov 3.407:

The magnetic induction in a betatron on an equilibrium orbit of radius r varies during the acceleration time at practically constant rate from zero to B. Assuming the initial velocity of the electron to be equal to zero, find:

(a) the energy acquired by the electron during the acceleration time;

(b) the corresponding distance covered by the electron if the acceleration time is equal to Δt.

Solution: (a) Refer to the triangle (relativistic calculations), $K = \sqrt{(m_0c^2)^2 + p^2c^2} - m_0c^2$

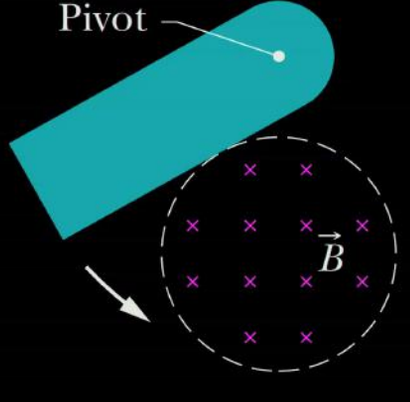
Also $R = mv/qB = p/qB$, thus, acquired energy:

$$K = \sqrt{m_0^2c^4 + (eBrc)^2} - m_0c^2 = m_0c^2\left(\sqrt{1 + (Ber/m_0c)^2} - 1\right)$$

(b) Now $K = N \times e\dot{\phi} = \frac{s}{2\pi r} \times e\left(\frac{2B}{\Delta t}\right) \times \pi r^2 \Rightarrow s = \frac{K\Delta t}{Ber}$

→ Note we used $2B$ in ϕ because B is given at periphery).

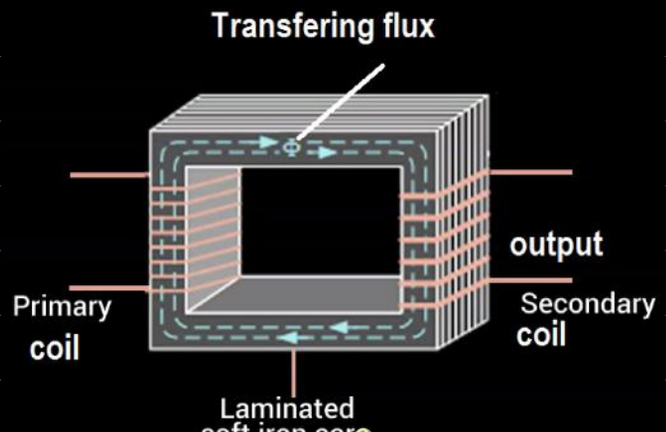
Eddy Currents: When metallic objects move in magnetic fields or are subject to time varying fields, current paths develop in them due to induced EMFs, these currents are called Eddy currents. The eddy currents lead to heat dissipation due to i^2R heating, thus a metallic rod oscillating in a magnetic field will lose



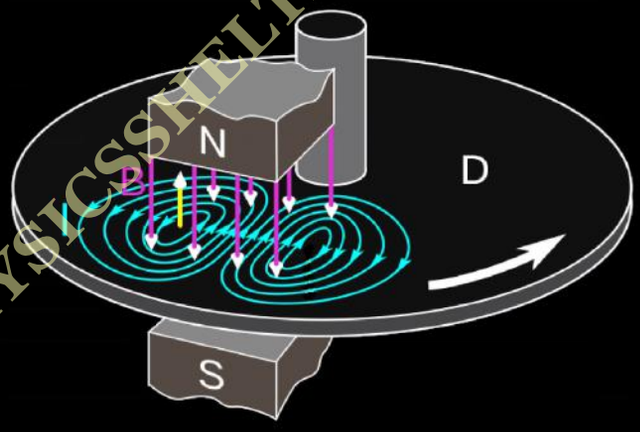
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kinetic energy due to eddy currents. In Solenoid and transformer cores, to avoid eddy losses, laminated cores are used, laminations cut out the current paths thus preventing i^2R losses.

→ A transformer is used to step up or step down voltages (schematic shown in adjacent figure). Transformer works on principle of electromagnetic induction.



→ Eddy Currents can also be used for applying brakes on rotating metal discs. (See figure). Here due to presence of magnetic field, eddie currents are generated which gives a back torque in accordance with Lenz's law.



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EXAMPLE : PATHFINDER MCQ 3:

In a cylindrical region that is encircled by a uniform metallic ring, a uniform magnetic field perpendicular to the plane of the ring varies with time in such a manner that a constant electromotive force of 12 V is induced in the ring. A voltmeter of very high resistance is connected between two points on the ring separated by one quarter of ring. In two different trials, the voltmeter is connected in two different manners as shown in figures. In both the arrangements, the ring and connecting wires are in the same plane. What conclusion can you draw regarding readings V_I and V_{II} of the voltmeter in the first and the second arrangement respectively?

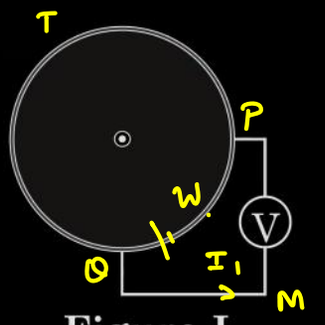


Figure-I

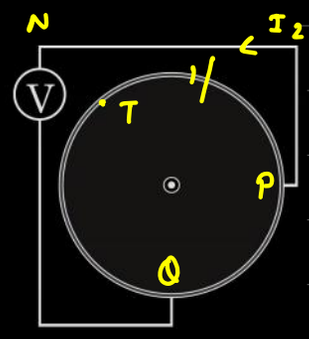


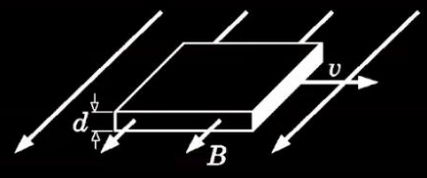
Figure-II

Solution: Since total EMF in the ring is 12 Volt, by symmetry, EMF in PWQ = 3 Volt and EMF in PNQ = 9 volt. Now let the resistance of the ring be 4R (R per each quarter) then, current in the ring $i = \mathcal{E}/4R$ where $\mathcal{E} = 12$ Volts. Now let i_v be the voltmeter current and R_v the voltmeter resistance then applying loop law to loop PQMP $-\mathcal{E}/4 + i_v R_v = 0$ ($\because \phi$ for this loop is zero), thus $i_v R_v = \mathcal{E}/4 = 3$ Volts ohm. Similarly for loop PNQP,

$-\frac{3\mathcal{E}}{4} + i_v R_v = 0 \Rightarrow i_v R_v = 3\mathcal{E}/4 = 9$ volt ohm. Note that this is not the potential difference (P.D. is zero).

Example: Pathfinder: Build 2:

A bar of thickness d is made to move with a velocity v perpendicular to a uniform magnetic field of induction B. Top and bottom faces of the bar are parallel to the lines of magnetic field as shown in the figure.



(a) Find surface charge densities on the upper and the bottom faces of the bar, assuming material of the bar to be a perfect conductor.

(b) Find the surface charge densities on the upper and the bottom faces of the bar, assuming material of the bar to be a dielectric of dielectric constant ε_r.

Solution: (a) The motion of the bar produces $\vec{E}^* = \vec{v} \times \vec{B}$ i.e a downward $E^* = vB$. From dynamics point of view this is as good as immersing the bar between parallel plates of a capacitor having charge density σ such that $\frac{\sigma}{\epsilon_0} = E^* = vB \Rightarrow \sigma = vB\epsilon_0$. Now with a metallic conductor, an equal and opposite charge density must develop on it faces to make the net field zero inside, thus, for conductor also, $\sigma = vB\epsilon_0$ Ans.

(b) Somewhat similar to the previous case, this is equivalent to immersing the dielectric in a parallel plate capacitor such that $\sigma/\epsilon_0 = vB$, now recall that surface density on a dielectric is given as $\sigma' = \sigma(1 - 1/k)$ thus:

$$\sigma' = vB\epsilon_0 [1 - 1/\epsilon_r] \text{ Ans.}$$

→ Note that here net electrostatic field in the conductor is not zero, rather $\vec{E} + \vec{E}^* = 0$.

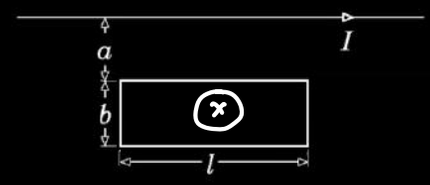
→ Alternatively we could do this problem by field transformation. We could go in the bar frame and use:

$\vec{E}' = \vec{E} + \vec{v} \times \vec{B}$ and in such a frame there will appear to be a real external applied electric field.

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EXAMPLE: PATHFINDER: BUILD-12:

A rectangular loop of width b, length l, mass m and resistance R is placed close to and coplanar with a very long fixed straight conductor carrying current I. One side of the loop is parallel to the straight conductor as shown in the figure. The current in the conductor is suddenly switched off. Neglecting gravity find velocity acquired by the loop.



Solution: To analyse the problem correctly we need to carefully focus on the very small time duration in which fields are changing suddenly. Instead of imagining the process to complete in zero time assume it to complete in some small amount of time and then imagine that time to be infinitesimal.

Now let the instantaneous current in the wire be I $\Rightarrow$

$$\phi = \frac{\mu_0 I l}{2\pi} \ln\left(\frac{b+a}{a}\right) \Rightarrow \dot{\phi} = \frac{\mu_0 l}{2\pi} \ln\left(\frac{b+a}{a}\right) \dot{I}$$

Now current in the rectangular frame, $i = \dot{\phi}/R$, By Lenz law the induced current causes a net attractive force between wire and the loop,

$$F = i l \left[\frac{\mu_0 I}{2\pi a} - \frac{\mu_0 I}{2\pi(a+b)}\right] = \frac{\mu_0 I i l}{2\pi}\left[\frac{1}{a} - \frac{1}{a+b}\right]$$

Thus $$F = \frac{\mu_0 l}{2\pi}\left[\frac{1}{a} - \frac{1}{a+b}\right] I \times \frac{1}{R}\left[\frac{\mu_0 l}{2\pi} \ln\left(\frac{b+a}{a}\right)\right] \dot{I}$$

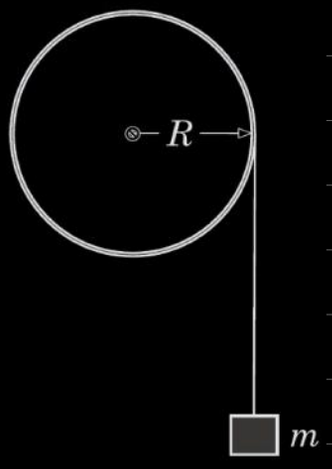
Now velocity of frame: $$v = \frac{1}{m}\int F dt = \frac{1}{m}\left(\frac{\mu_0 l}{2\pi}\right)^2 \cdot \frac{1}{R} \frac{b}{a(b+a)} \ln\left(1+\frac{b}{a}\right) \cdot \frac{I^2}{2}$$

Thus $$v = \frac{\mu_0^2 b l^2 I^2}{8\pi^2 m R a(a+b)} \ln\left(\frac{a+b}{b}\right) \text{ Ans.}$$

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Example: Pathfinder: Build-16

A uniform dielectric hollow cylinder of mass M, radius R, length l carrying uniform charge of surface charge density σ can rotate without friction about a fixed horizontal axle that coincides with the axis of the cylinder. Several turns of a light thin insulating cord are wrapped on the cylinder and a block of mass m is suspended from the free end of the cord. Initially the block is held at rest as shown in the figure. Find acceleration of the block after it is released. Neglect charge transferred to the cord and fringing of magnetic field at the ends of the cylinder. Acceleration due to gravity is g and permeability of the medium inside the cylinder is μ_0.



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Solution: Here rotation of the cylinder produces a time varying magnetic field which produces E^* which in turn provides a reverse torque due to force on surface charge density.

Let the acceleration of the block be $a \Rightarrow \alpha = a/R$. Now magnetic field inside cylinder, $B = \mu_0 \times \frac{\sigma \times 2\pi R \times l}{(2\pi/\omega) l} = \mu_0 \sigma \omega R$

Now $E^* = \dot{B} \times \pi R^2 / 2\pi R = \dot{B} R/2 = \mu_0 \sigma \alpha R^2/2$, thus,

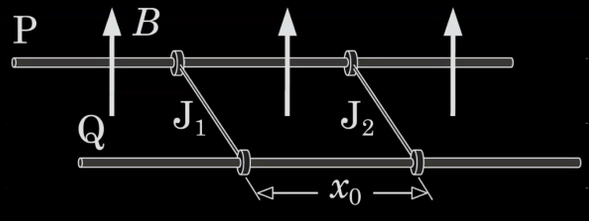
Torque due to E^*, $\tau^* = (\sigma \times 2\pi R l) \times E^* \times R = \pi \sigma^2 R^4 l \alpha \mu_0$

Now Net torque about the pivot $= dL/dt$ of the system, thus:

$$\left[mgR - \mu_0 \pi \sigma^2 R^4 l \alpha\right] = mR(\alpha R) + MR^2\alpha \Rightarrow \alpha R = \frac{mg}{m + M + \sigma^2 \pi R^2 l \mu_0}$$ Ans.

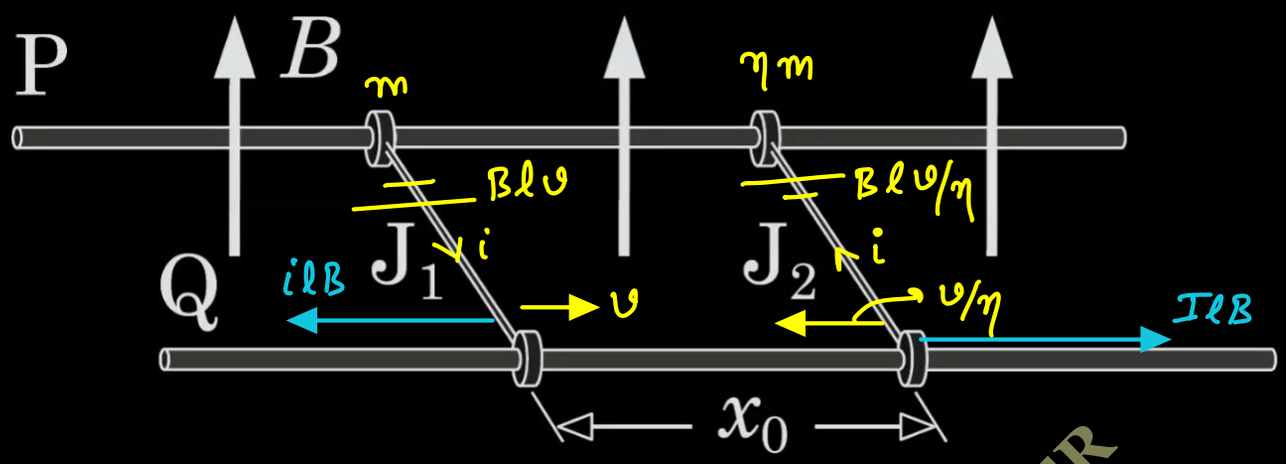
Example: Pathfinder: Check 1:

Two long conducting bars P and Q are fixed parallel to each other in a horizontal plane some distance apart. On these bars, two parallel jumpers J_1 and J_2 each of resistance R are arranged at separation x_0 as shown in the figure. The jumpers can slide on the bars without friction. If a uniform vertical magnetic field of induction B is switched on, find final separation between the jumpers. Force between the jumpers due to their currents is negligible as compared to the force of interaction between current in a jumper and the magnetic field.



Solution: Here due to sudden change in flux, there will be a sudden charge flow causing an impulsive magnetic force and hence a velocity in jumpers. After this both jumpers will experience

a retarding magnetic force due to induced currents.



Here we should carefully focus on switching process. Instead of assuming magnetic field to change in zero time imagine it to be changing in a very small amount of time. During this time, the current in the jumpers,

$i = \frac{1}{2R}\frac{d\phi}{dt} = \frac{1}{2R}\ell x_0\left(\frac{dB}{dt}\right)$, Now force on each jumper,

$F = i\ell B = \frac{1}{2R}\ell x_0 \frac{dB}{dt}\cdot \ell B \Rightarrow J = \int F dt = \frac{\ell^2 x_0}{2R}\frac{B^2}{2}$

$\Rightarrow J = \frac{B^2\ell^2 x_0}{4R}$, thus velocity of left jumper, $v_0 = \frac{B^2\ell^2 x_0}{4mR}$

similarly for right jumper, velocity = v_0/η (since mass is η times).

[Without loss of generality we assumed jumpers of masses m and ηm].

Now let at any arbitrary instant velocities of jumpers be v and v/η then induced current:

$$i = B\ell\left[v + \frac{v}{\eta}\right]\cdot\frac{1}{2R}$$

Now acceleration of left jumper, $a = \frac{-i\ell B}{m} = v\frac{dv}{ds} = -\frac{B^2\ell^2 v}{2mR}\left[1+\frac{1}{\eta}\right]$

Now @ $s=0$, $v=v_0$ and at $s=s_f$, $v=0$ where s_f is final displacement, thus $s_f = \frac{2mv_0R}{B^2\ell^2}(\eta/\eta+1)$

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Now displacement of the other jumper $s_f' = s_f/\eta$ ($\because$ at every instant, its velocity is $1/\eta$ times).

Thus total displacement, $s = s_f + s_f' = \frac{2mv_0R}{B^2l^2} = x_0/2$

thus final separation between jumpers, $x = x_0 - x_0/2 = x_0/2$ Ans.

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